



THE BEGINNER'S GUIDE

# Network Protocols **EXPLAINED**

How the internet actually talks, from zero

SWIPE TO LEARN



# What is a protocol?



## A shared set of rules

An agreed language two devices use to exchange data. It sets the format, order, and meaning of every message.

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*“Like two people agreeing to speak the same language before they start talking.”*

## WHY RULES ARE NEEDED



### Same language

Both sides interpret the raw bytes in the exact same way, with no ambiguity.



### Right order

Each step happens in a known sequence, so neither side has to guess what comes next.



### No chaos

Millions of devices from different makers interoperate smoothly, without conflict.

No single protocol does everything. Each handles one job and stacks on the one below it.



# IP: the addressing layer

Every device gets an IP address. Data is split into packets, each stamped with where it is going.

## ANATOMY OF A PACKET

## HEADER

## from + to address

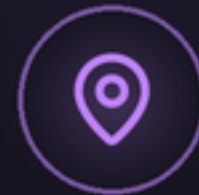
## PAYLOAD

the actual data



## Routers forward it

Each router reads the destination address and passes the packet one hop closer, like a chain of postal sorting offices.



## Two versions

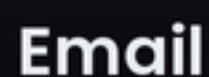
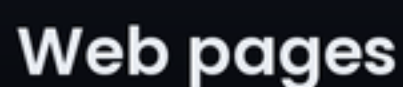
**IPv4** older, limited supply

IPv6 newer, vastly larger

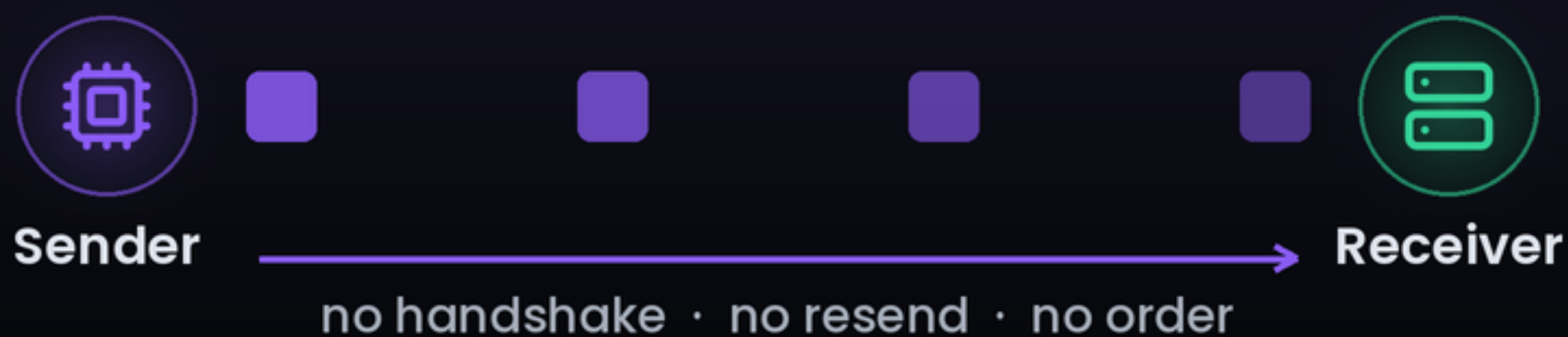


Big data is split into many numbered packets, then reassembled in the right order once they all arrive.

TCP sets up a connection first, then guarantees every piece arrives, in order, with nothing lost.



UDP skips the setup and the guarantees. It just fires packets out, fast, and never looks back.



## DNS lookups

# TCP vs UDP

Same job, opposite trade-offs. One prizes accuracy, the other prizes speed.

|             | TCP          | UDP              |
|-------------|--------------|------------------|
| Connection  | Set up first | None             |
| Reliability | Guaranteed   | Best effort      |
| Ordering    | In sequence  | May arrive mixed |
| Speed       | Slower       | Very fast        |
| Overhead    | Higher       | Minimal          |
| Best for    | Accuracy     | Real-time        |

# DNS: the phonebook

People remember names, computers need numbers. DNS translates a website name into its IP address.

**a-website-name**

easy for humans



**203.0.113.5**

what machines use

## HOW A LOOKUP WORKS

**1**

**You type a name**

The browser needs its IP to connect.

**2**

**Ask a resolver**

A DNS server is queried on your behalf.

**3**

**Get the answer**

The matching IP address comes back.

**4**

**Connect**

Your browser reaches the real server.



# DHCP: auto addressing

When a device joins a network it has no address yet. DHCP hands it one automatically, in seconds.



Think of a hotel front desk: you arrive, it assigns you a room number, gateway, and DNS, all at once.

## THE DORA EXCHANGE



### Discover

Device shouts: anyone manage addresses?



### Offer

A server replies with a free address.



### Request

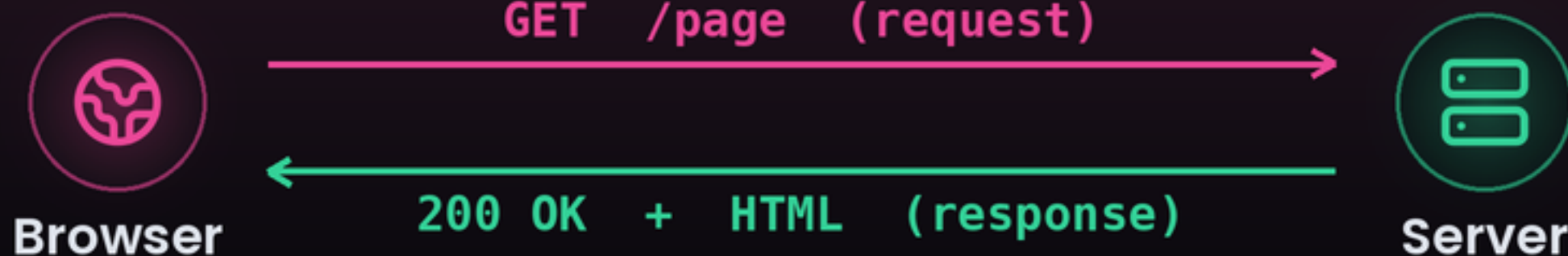
Device asks to keep that one.



### Acknowledge

Server confirms. Device is online.

The web runs on a simple loop: your browser sends a request, the server sends a response back.



- GET** read data
- POST** send data
- PUT** update data
- DELETE** remove data



HTTPS is plain HTTP wrapped in encryption. Nobody between you and the server can read or alter the data.

## Success

## Redirect

## Client error

Server error

# TLS: the lock layer

TLS (older name SSL) is the padlock in your address bar. It secures data while it travels.



scrambled in transit, readable only at the ends

## WHAT IT GUARANTEES



## Privacy

Outsiders cannot read your data.



## Integrity

Data cannot be secretly changed.



## Identity

**You reach the real server, not a fake.**

# Email protocols

Email uses different protocols for sending versus reading. Knowing which is which clears up a lot.



## **SMTP** Sends mail

Pushes your message out to the recipient's mail server.



## **IMAP** Reads on server

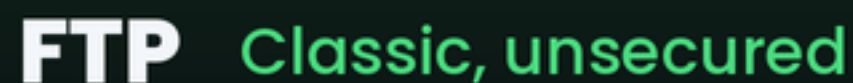
Keeps mail on the server and syncs it across all your devices.



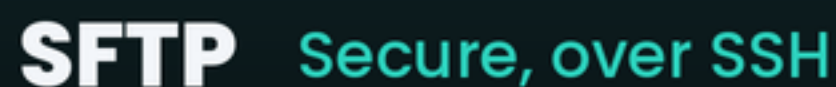
## **POP3** Downloads mail

Pulls messages onto one device, often removing them from the server.

Several protocols move files between machines. They differ mainly in how secure they are.



Transfers files but sends them in the clear. Avoid for anything private.

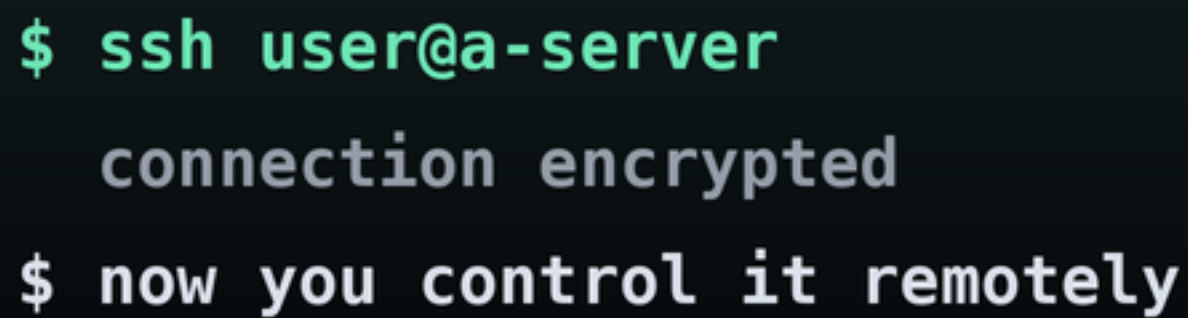


Encrypts the whole transfer. The safe default choice today.

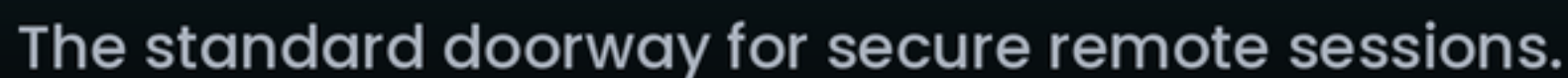
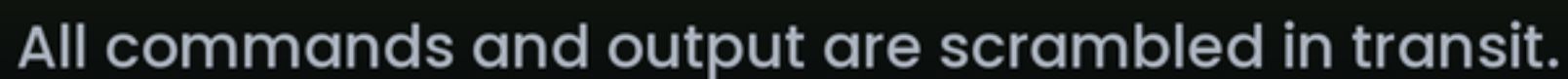


Bare-bones, used inside trusted networks for things like device boot.

SSH lets you safely control another computer over the network, with everything encrypted end to end.



Logs in with cryptographic keys, stronger than passwords.



Some protocols never carry your content. They quietly keep the network working behind the scenes.



Powers tools like ping and traceroute, and reports when a destination is unreachable.

## error reports



Maps an IP address to a physical (MAC) address so devices can reach each other on the same network.

MAC

A port is a numbered door on a device. Each protocol listens on its own well-known one.

 @kerem.tech  @keraattin 



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